

7 Planar and Non-planar Graphs

A graph is a **planar graph** if it can be drawn in the plane with no edge crossings. [Note: You may need to rearrange the vertices and edges to avoid crossings.]

7.1 Euler and Planarity

We have already seen **Euler's Formula for Planar Graphs**:

$$v - e + r = 1 + \text{number of connected components},$$

where v is the number of vertices, e the number of edges and r the number of regions.

In particular, for a connected graph we have $v - e + r = 2$.

1. Show that the following are planar.
 - a. $K_{2,2}$
 - b. $K_{2,3}$
 - c. $K_{2,4}$
2. Is $K_{2,n}$ planar for every n ?
3. Use Euler's formula to show that K_5 is not planar.
 - a. How many vertices does K_5 have?
 - b. How many edges does K_5 have?
 - c. If K_5 were planar, how many regions would it have?
 - d. Show that K_5 can't have that number of regions, so it must not be planar.
4. Use Euler's formula to show that $K_{3,3}$ is not planar.

7.2 Kuratowski's Theorem

The proof of the following theorem is too involved for this course, but it turns out that all nonplanar graphs contain a “homeomorphic copy” of either K_5 or $K_{3,3}$.

Kuratowski's Theorem: A graph G is nonplanar if and only if it contains a subgraph H that is homeomorphic to either K_5 or $K_{3,3}$.

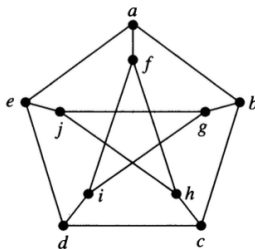
- an **elementary subdivision** of a graph G removes one edge (u, v) and adds a new vertex n and two new edges (u, n) and (v, n) .
 - a **subdivision** of a graph G is the result of performing 0 or more elementary subdivisions, starting from G .
 - two graphs are **homeomorphic** if some subdivision of one is isomorphic to some subdivision of the other.
5. Create a graph with 5 vertices and 7 edges. Now perform three elementary subdivisions on your graph.
 6. Describe what an elementary subdivision is in simple words a kindergartner could understand.
 7. Explain why two homeomorphic graphs are either both planar or both nonplanar.
 8. Use Kuratowski's Theorem to show that the Petersen graph is nonplanar.

There is more than one way to do this, but if you are having trouble, here are some things you could try.

- a. First, decide whether you are looking for a version of K_5 or a version of $K_{3,3}$ inside the Petersen graph. (It's pretty easy to give an argument that one of these two will not work. Identifying which graph to work with will save you going down a wrong path right from the start.)
- b. Now see if you can identify
 - the 5 or 6 vertices in the Petersen graph that will correspond to the vertices in K_5 or $K_{3,3}$.
 - any vertices or edges that you will remove from the Petersen graph. (We only need to work with a subgraph, so we are allowed to remove things if they get in the way.)
 - any vertices in the Petersen graph that will remain, but not be one of the vertices corresponding to K_5 or $K_{3,3}$. These must be the vertices involved in subdivisions.

There may be some trial and error involved here.

- c. When you think have identified everything correctly, double check to make sure the you have the correct structure for K_5 or $K_{3,3}$.



9. Show that the Petersen graph is not planar *without* using Kuratowski's Theorem.